

GED Math: Algebra Basics, Expressions, and Polynomials

GED Preparation Course

What is Algebra?

- Algebra uses symbols (letters) to represent numbers.
- Symbols are called **variables**.
- Example: $x + 5 = 9$.

Key Concepts

- **Variable:** A symbol (e.g., x , y) representing an unknown.
- **Constant:** A fixed number (e.g., 3, -7, 12).
- **Coefficient:** A number multiplying a variable.
- **Equation:** A mathematical statement of equality.

Worked Examples

Example 1: Solve $x + 7 = 12$.

$$x = 12 - 7 = 5$$

Example 2: Solve $-3x = 12$.

$$x = \frac{12}{-3} = -4$$

Example 3: Solve $3x + 4 = 19$.

$$x = 5$$

Example 4: Solve $\frac{x}{4} = 3$.

$$x = 12$$

Example 5: If $5x = 25$, find x .

$$x = 5$$

What are Algebraic Expressions?

- An **expression** combines variables, constants, and operations.
- Example: $3x + 2y - 5$.
- No equals sign.

Worked Examples

Example 1: Simplify $4x + 3x - 2 = 7x - 2$.

Example 2: Simplify $2(3x + 4) = 6x + 8$.

Example 3: Simplify $-2(4x - 5) = -8x + 10$.

Example 4: Simplify $5x + 2y - 3x + 4y = 2x + 6y$.

Example 5: Evaluate $2x + 3$ when $x = 4$.

$$= 2(4) + 3 = 11$$

What is a Polynomial?

- A **polynomial** has terms with variables raised to whole number powers.
- Example: $3x^2 + 2x + 1$.
- The **degree** is the highest power.

Worked Examples

Example 1: Add $(3x + 2) + (4x + 5) = 7x + 7$.

Example 2: Subtract $(6x^2 + 3x) - (2x^2 + 5x) = 4x^2 - 2x$.

Example 3: Multiply $3x(2x + 4) = 6x^2 + 12x$.

Example 4: Expand $(x + 2)(x + 3) = x^2 + 5x + 6$.

Example 5: Factor $x^2 + 5x + 6 = (x + 2)(x + 3)$.

Lesson 1 Practice: Algebra Basics

- 1 Solve $x - 9 = 15$.
- 2 Solve $5x = 45$.
- 3 Solve $x/3 + 4 = 9$.
- 4 Solve $2x - 5 = 11$.
- 5 Solve $7x + 3 = 31$.

Lesson 2 Practice: Expressions

- 1 Simplify $2x + 3x + 4$.
- 2 Simplify $3(2x + 5)$.
- 3 Simplify $4y - 2(3y - 6)$.
- 4 Simplify $5x + 4y - x + 2y$.
- 5 Evaluate $3x + 2$ when $x = 6$.

Lesson 3 Practice: Polynomials

- 1 Add $(2x^2 + 3x + 1) + (x^2 + 4x + 5)$.
- 2 Subtract $(5x^2 + 6x) - (3x^2 + 2x)$.
- 3 Multiply $2x(3x + 1)$.
- 4 Expand $(x + 4)(x + 2)$.
- 5 Factor $x^2 + 7x + 10$.

Lesson 1 Answers

① $x = 24$

② $x = 9$

③ $x = 15$

④ $x = 8$

⑤ $x = 4$

Lesson 2 Answers

① $5x + 4$

② $6x + 15$

③ $4y - 6y + 12 = -2y + 12$

④ $4x + 6y$

⑤ 20

Lesson 3 Answers

① $3x^2 + 7x + 6$

② $2x^2 + 4x$

③ $6x^2 + 2x$

④ $x^2 + 6x + 8$

⑤ $(x + 5)(x + 2)$

Summary

- Algebra helps solve for unknowns using symbols.
- Simplify expressions by combining like terms and using distribution.
- Polynomials follow consistent rules for addition, subtraction, multiplication, and factoring.